

Report on NAC Breda

Enhancing Attacker Acquisition Strategy at NAC Breda: Ethical Frameworks and Data-Driven Insights



DISCOVER YOUR WORLD

Report on NAC Breda

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1 Introduction

In the competitive realm of professional football, strategic player acquisition is pivotal for staying ahead. This report tackles the business challenge presented by NAC Breda by leveraging data-driven insights to enhance their player recruitment strategy. Through Exploratory Data Analysis on player data from the 2022 and 2023 seasons, a sophisticated tool fuelled by machine learning models was developed to address this need.

Further analysis revealed that contracts for 11 players, including 4 attackers, were set to expire on June 30, 2024, creating a crucial gap for NAC Breda.¹ Consequently, the focus shifted to identifying suitable replacements for attackers, aiming to craft a tool that aided informed recruitment decisions based on tailored scoring metrics.

Ethical considerations were central to the approach, with all processes and criteria aligning with diverse ethical frameworks. An evaluation of NAC Breda's ethical capacity, addressing potential challenges, and providing recommendations for GDPR and Ethical Guidelines for Statistical Practice compliance was undertaken. This approach optimized player acquisition strategies while promoting ethical and responsible data practices for NAC Breda.²

2 Exploratory Data Analysis

2.1 Data Overview

The preliminary step in the exploratory data analysis process involves understanding the dataset. Supplied by our client, the data comprises 45 Excel sheets, each documenting players from different European leagues and countries for the 2022-2023 season.

	Player	Team	Team within selected timeframe	Position	Age	Market value	Contract expires	Matches played	Minutes played	Goals	...	Prevented goals per 90	Back passes received as GK per 90	Exits per 90	Aerial duels per 90.1	Free kicks per 90
0	P. Iemmello	Catanzaro	Catanzaro	CF	31.0	750000	NaN	38	3109	30	...	NaN	0.00	NaN	NaN	0.00
1	J. Petriccione	Crotone	Crotone	DMF, RDMF, RCMF	28.0	700000	2024-06-30	36	3341	2	...	NaN	0.32	NaN	NaN	1.29
2	T. Biasci	Catanzaro	Catanzaro	CF, LWF, AMF	28.0	550000	2024-06-30	38	2488	17	...	NaN	0.07	NaN	NaN	0.00
3	E. Volpicelli	Sangiuliano City	Sangiuliano City	CF, RWF, AMF	30.0	500000	2024-06-30	34	2639	3	...	NaN	0.10	NaN	NaN	2.28
4	A. Curcio	Catanzaro	Catanzaro	CF, AMF, LWF	33.0	500000	2024-06-30	38	1456	9	...	NaN	0.00	NaN	NaN	0.62

Figure 1. The head of our football player dataset

After combining multiple data sheets, the football player dataset comprises 16,535 rows and 114 columns. Each row represents a player, with columns containing personal, biometric, and scoring features.

Some of the scoring statistics incorporated in the dataset are more pertinent to specific positions. Given our project's focus on the attacker position, the most pertinent scoring statistics for this role include Goals, xG, Assists, xA, Duels per 90, Successful attacking actions per 90, Shots, Shots on target %, Goal conversion %, Crosses per 90, Dribbles per 90 and many others. ²

The dataset includes 9 player-related columns and 105 numerical columns, mostly containing decimal numbers representing scoring statistics. Some missing values, evidenced by "NaN", require data cleaning before exploratory data analysis to enhance data quality and ensure reliability for subsequent analyses.²

2.2 Data Cleaning and Pre-processing

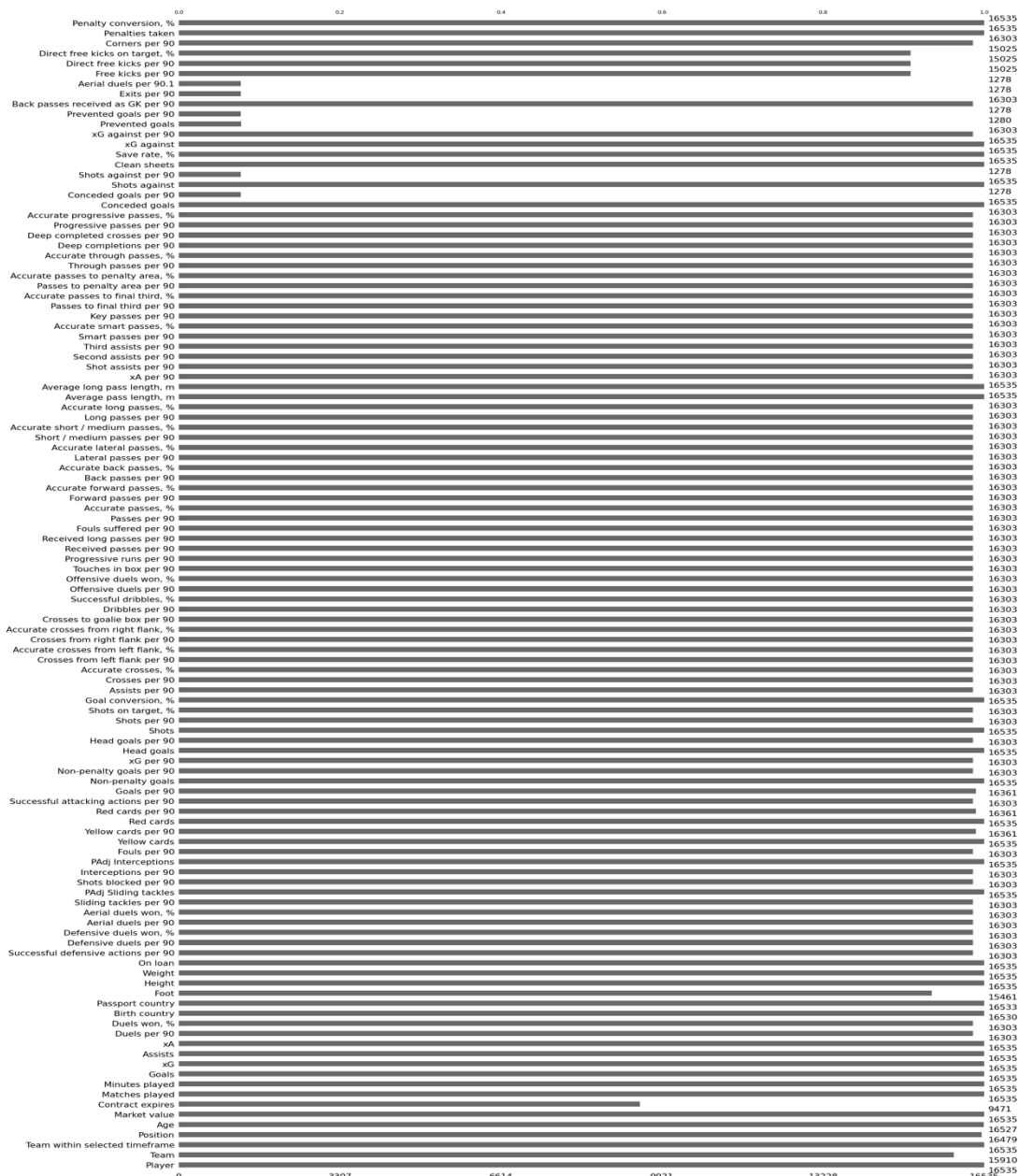


Figure 2. Bar chart with missing data

2.2.1 Handling missing data

The initial step in data set cleansing involves identifying and addressing missing data. The strategy for handling missing values often depends on the type of missingness and the nature of the data. The bar chart above reveals that the majority of columns have over 95% of available data.

For handling missing data in critical features, the following strategy has been chosen. Columns such as Position, Age, Birth country, and Passport country, with a limited number of missing entries specific to

each player, were addressed by dropping the corresponding rows. For the Foot column, replacing missing values with "Unknown" provided a sensible approach. Similarly, in the Contract expires column, missing data were filled with "Unknown", as the contract end dates are non-essential for the analysis. Concerning the Team column, players not associated with any team can be denoted as "Free Agents."²

A further investigation was conducted into features with a high number of missing data to understand the type and nature of the missingness. It was found that the missing data in certain columns, were not missing at random, but rather connected to a specific football position. Therefore, these missing values were replaced with 0 to maintain data integrity and consistency. Additionally, statistical features with minimal missing data were also filled with 0, possibly indicating young players from youth teams who haven't participated in matches but have these features recorded.²

2.2.2 Initial Pre-processing

After handling missing values, the dataset underwent preprocessing steps to ensure accuracy and consistency. Duplicate entries were removed by considering player name, team, and age, resulting in a clean dataset of 113 columns and 16,220 rows. To facilitate analysis focusing on attackers, the 'Position' column was expanded to include individual positions, and players were categorized into broader "Position categories" such as attackers, defenders, midfielders, and goalkeepers. Furthermore, a "Bin" feature was introduced to classify players into three market value bins. Finally, datatype inspection revealed that the "Contract expires" column needed conversion to the date format to accurately represent contract end dates.²

2.2.3 Further Pre-processing

For the dataset to closely resemble the European football scene in the EDA, outliers remained unhandled at this stage. Further preprocessing was conducted before building the machine learning model. Players' market values were capped at 350,000 euros based on NAC Breda's average value from Transfermarkt, aligning with the club's financial range.²

Columns specific to individual players without significant predictive value, such as "Player," "Team," "Team within selected timeframe," "Position," "Contract expires," and race-sensitive features like "Birth

country" or "Passport country," were dropped, ensuring a non-racial approach. Remaining categorical data like "On loan," "Foot," and "Bin" were one-hot encoded. A new subset, "attackers," was created, consisting only of players with the position category "ATT," indicating our focus position. Before inputting data into the machine learning tool, all input data was scaled.

2.3 Summary Statistics

To gain further insight into the data, a distribution analysis was conducted, and summary statistics were computed. The analysis revealed that the numerical data did not follow a normal distribution and were instead skewed. Most of the data exhibited right skewness, possibly due to the replacement of missing values with 0, while some displayed left skewness.

	Age	Market value	Contract expires	Matches played	Minutes played	Goals	xG	Assists	xA
count	16220.000000	1.622000e+04	16220	16220.000000	16220.000000	16220.000000	16220.000000	16220.000000	16220.000000
mean	25.264365	5.075937e+05	2001-06-08 15:33:36.177558656	20.998397	1496.407090	1.810173	1.908044	1.066708	1.130173
min	15.000000	0.000000e+00	1970-01-01 00:00:00	1.000000	1.000000	0.000000	0.000000	0.000000	0.000000
25%	22.000000	1.000000e+05	1970-01-01 00:00:00	14.000000	762.000000	0.000000	0.300000	0.000000	0.130000
50%	25.000000	2.250000e+05	2024-05-31 00:00:00	21.000000	1404.000000	1.000000	1.020000	0.000000	0.640000
75%	29.000000	4.500000e+05	2024-12-31 00:00:00	28.000000	2130.250000	2.000000	2.430000	2.000000	1.580000
max	42.000000	6.000000e+07	2032-06-30 00:00:00	45.000000	4491.000000	30.000000	23.230000	22.000000	21.510000
std	4.637327	1.405688e+06	NaN	9.172757	863.329721	2.848828	2.554669	1.621991	1.436532

Figure 4. Summary statistics (Part I)

Duels per 90	Duels won, %	Height	Weight	Successful defensive actions per 90	Defensive duels per 90	Defensive duels won, %	Aerial duels per 90	Aerial duels won, %
16220.000000	16220.000000	16220.000000	16220.000000	16220.000000	16220.000000	16220.000000	16220.000000	16220.000000
17.676594	50.343017	160.002466	62.763564	7.237658	5.626409	57.626953	3.309841	45.579906
0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
14.320000	41.997500	175.000000	67.000000	4.770000	4.230000	53.260000	1.760000	33.330000
17.980000	49.780000	181.000000	74.000000	7.650000	5.850000	59.570000	2.830000	44.750000
22.130000	57.140000	186.000000	79.000000	9.620000	7.330000	65.570000	4.340000	56.250000
62.000000	100.000000	206.000000	103.000000	20.500000	18.000000	100.000000	21.060000	100.000000
7.444003	14.173872	59.911651	28.642108	3.217080	2.618186	16.290511	2.350766	20.266750

Figure 3. Summary statistics (Part II)

The summary statistics provided a comprehensive overview of the dataset. The average age of players was 25 years, with a wide range from 15 to 42 years old. The average market value was approximately 50,760 euros, with notable variability. Height and weight averages were skewed by younger players, but median values of 181 cm and 74 kg better represented adults. The highest number of goals scored was 30, with an average of less than 2 goals per player. Players participated in around 20 matches and played approximately 1496 minutes on average. Categorical variables showed diversity, with players representing 1678 teams and 155 countries of birth. Italy and Germany were most represented. Right-footed players were predominant, with defenders being the most common position. The "Medium" market value bin was the most frequent, with around 6,000 players.²

2.4 Visualisation Techniques and Insight from Data

Various visualization techniques, including histograms, scatterplots, bar charts, pie charts, box plots, and line charts, were employed during the exploratory data analysis (EDA) to gain insights into the dataset.

The analysis revealed that teams typically have 3 to 4 attackers, with each averaging 1 to 5 goals per season, although some achieve higher counts, up to 12 goals. Most attackers prefer their right foot, with a minority favouring their left. Top-performing attackers are typically aged between 20 and 25, exhibiting high goal counts and successful attacking actions, and command higher salaries.²

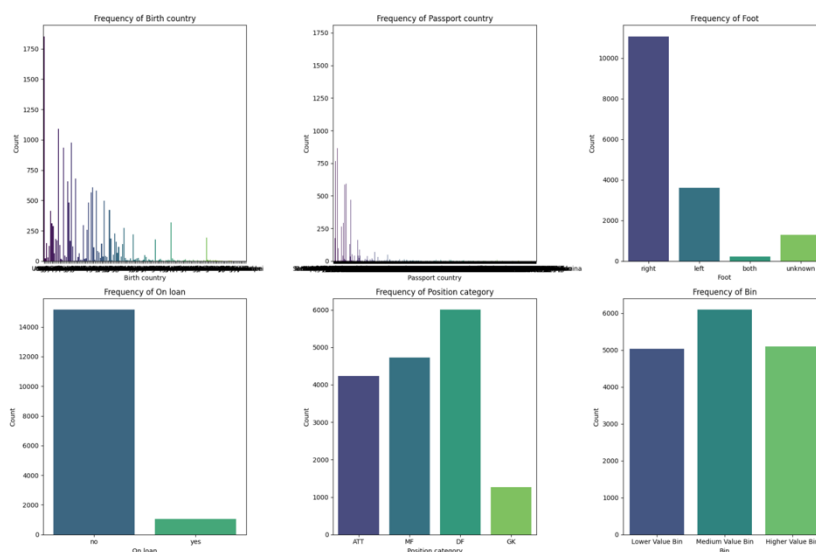


Figure 5. Frequency counts for categorical variables

Weak positive correlations were observed between various scoring features, such as “Shots on target” and “Goal conversion”. However, a strong positive correlation was found between “Dribbles per 90” and “Successful attacking actions per 90”. However, a strong positive correlation was found between “Dribbles per 90” and “Successful attacking actions per 90”, suggesting that higher dribble rates directly influence successful attacking actions. Leveraging this correlation could aid NAC Breda in their attacker acquisition strategy, although further analysis is needed to explore its depth and connection to other relevant features.²

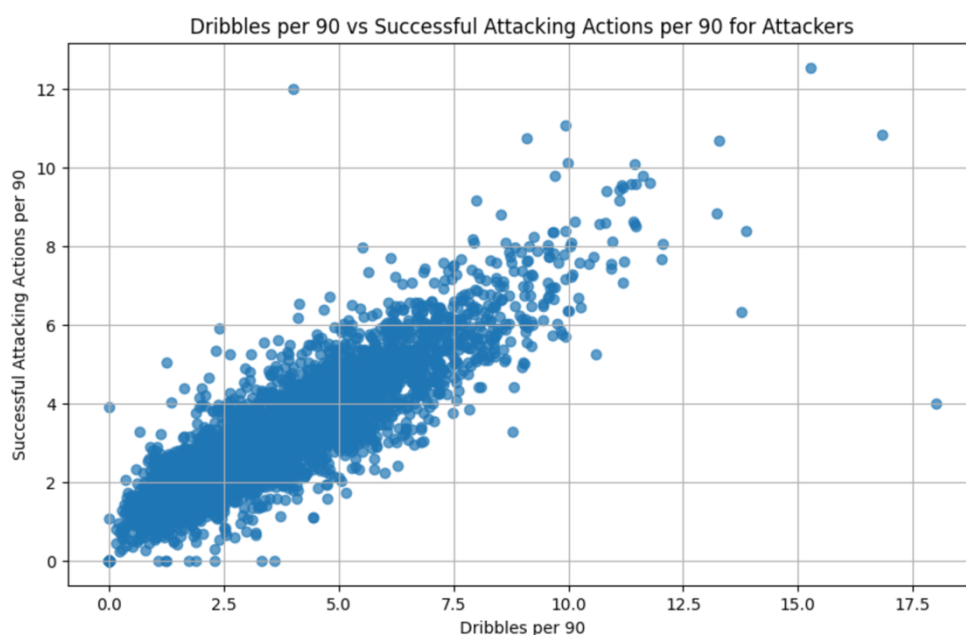


Figure 6. Scatter plot for positive correlation of “Dribbles per 90” and “Successful attacking actions per 90”

2.5 Correlation Analysis and Feature Selection

The project's main objective is to provide NAC with a tool for effectively identifying suitable attackers for their team. In pursuit of this goal, “Successful attacking actions per 90” was identified as the key feature defining a successful attacker, subsequently becoming our target variable.

Leveraging a random forest regressor, it was determined that “Dribbles per 90” emerged as the most highly related feature to our target variable, contributing over 75% to its definition. This observation is consistent with the strong correlation noted during exploratory data analysis (EDA) between this feature

and the target variable. Additionally, "Successful dribbles, %" and "Crosses per 90" were identified as significant features, displaying notable correlations in the correlation matrix with both the target variable and "Dribbles per 90."²

Based on this correlation analysis, the aforementioned features were selected as predictor variables for the target variable.

2.6 Key Findings from EDA

The EDA revealed two key findings. Firstly, successful attackers typically range from 20 to 25 years old, indicating their significance in NAC Breda's acquisition strategy. Secondly, a notable linear relationship between "Dribbles per 90" and "Successful attacking actions per 90" was observed, confirming their relevance in defining proficient attackers. This led to the selection of "Successful attacking actions per 90" as a crucial metric. ²

One potential hypothesis arising from this analysis is that harnessing AI and machine learning algorithms could potentially create a robust tool to aid NAC Breda in attacker acquisition, driven by the strong positive correlation observed between "Dribbles per 90" and "Successful attacking actions per 90." These features appear pivotal in defining successful attackers and correlate significantly with "Successful dribbles %" and "Crosses per 90." ²

To enhance accuracy, selecting a model capable of handling complex relationships and exploring additional features for more comprehensive analysis are recommended. This approach could refine the model for optimal decision-making support and improve player performance assessments.²

3 Machine Learning

3.1 Method

Because our goal is to predict "Successful attacking actions per 90" based on our chosen predictor features regression models were employed for their compatibility with numerical variables. After testing various algorithms, the XGBoost regressor was chosen for its efficiency in handling structured data.

XGBoost algorithm sequentially builds an ensemble of weak learners, correcting errors made by previous ones. Its advantages include capturing complex relationships, effective handling of missing values, preventing overfitting, and managing non-linearity and feature interactions. This makes it ideal for identifying suitable attackers for NAC's team, as which involves understanding intricate relationships between different performance metrics.²

3.2 Model Evaluation

Various metrics were used to evaluate our machine learning model's performance, including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), R-squared (R²), and Mean Absolute Error (MAE). MSE and RMSE measured prediction accuracy by assessing the squared and square root of the differences between predicted and actual values. R² indicated the proportion of variance in the dependent variable predictable from independent variables. Additionally, MAE measured the average absolute difference between predicted and actual values, offering robust error insights.²

These metrics were chosen because they provide comprehensive insights into different aspects of the model's performance. MSE and RMSE quantify the accuracy of the predictions, R-squared assesses the overall goodness of fit, and MAE provides a measure of the average prediction error, making them suitable choices for evaluating the model across various dimensions.²

The untuned XGBoost regressor model with default parameters displayed inferior performance compared to other models tested. It exhibited an MSE of 0.39 and an RMSE of 0.63, indicating higher prediction error than alternatives, typically around MSE 0.30 and RMSE 0.55. Its R² value of 0.87 explained approximately 87% of the variance in the target variable, lower than other models achieving around 0.9. Similarly, its MAE of 0.45 exceeded the 0.40 achieved by other models.²

To ensure robust evaluation and mitigate overfitting concerns, k-fold cross-validation with 5 folds was utilized. The cross-validation R-squared scores ranged from 0.84 to 0.89 across different folds, with an average R-squared of 0.87, consistent but slightly lower than the 0.88 average achieved by other models.²

Despite the untuned model's unsatisfactory performance, hyperparameter tuning significantly improved its effectiveness, emerging as the top-performing model among all algorithms tested.

3.3 Model Improvement

During model tuning, several key hyperparameters like the number of estimators (`n_estimators`), maximum depth of trees (`max_depth`), and the learning rate (`learning_rate`) were optimized with 4 to 5 values each to enhance performance.

To systematically search for the best hyperparameters, GridSearchCV exhaustively tested all parameter combinations to determine the best ones, along with the best score. For the XGBoost regressor, optimal parameters were a learning rate of 0.1, max depth of 3, and 100 estimators.

Hyperparameter adjustments significantly improved model performance. After tuning, MSE decreased to 0.26, RMSE to 0.51, and R2 increased to 0.91, with MAE dropping to 0.38. Furthermore, cross-validation R-squared scores averaged 0.89, indicating better generalization. These results surpassed those of other fine-tuned models, establishing the XGBoost regressor as the preferred tool for NAC Breda's attacker acquisition strategy.²

4 Ethical Considerations

4.1 Three Essential Ethical Elements and Parties Responsible for Them

In striving to be an ethical organization, NAC Breda must adhere to three essential elements. Firstly, it must implement ethical policies for employees and stakeholders, prioritizing diversity, transparency, and feedback responsiveness. Secondly, ethics should be integral to decision-making and operations. Lastly, NAC Breda must promote ethical conduct among employees and clients, emphasizing integrity, honesty, and respect, while raising awareness of ethics and morals among stakeholders.²

Multiple stakeholders, including senior management, HR, and legal departments, oversee ethical practices, while data protection officers, legal teams, and IT personnel manage ethical processes and tools for data management at NAC Breda. Collaboration among these stakeholders ensures ethical behaviour among employees and clients, maintaining standards throughout the organisation.²

4.2 Insights about NAC Breda's Ethical Organizational Capacity

Research indicates that NAC Breda demonstrates a commitment to ethical practices across its operations, focusing on three core elements. Firstly, the organisation upholds ethical policies and behaviours towards employees and stakeholders, aligning with GDPR and Ethical Guidelines for Statistical Practice principles for data management. The club promotes diversity, maintains ethical relationships with sponsors, and fosters a culture of inclusivity and integrity. Secondly, NAC Breda integrates ethics into its processes and tools, exemplified by its privacy statement and adherence to ethical frameworks such as those outlined by the NAC Supporters Association and Club Council. These frameworks ensure compliance with legal and regulatory requirements, reinforcing the organisation's commitment to ethical conduct. Additionally, the club's terms and conditions outline responsible behaviour for fans during matches, further reinforcing its commitment to ethical standards. Lastly, NAC Breda promotes ethical behaviour among employees, sponsors, and fans through mechanisms like consent mechanisms for data management and ethical conduct guidelines.^{3, 4, 2, 5}

In our project, GDPR compliance was maintained by excluding individual names and race/ethnicity from the attacker dataset to protect personal information and avoid bias in machine learning algorithms. Biometric data such as weight or height was also omitted from the model to ensure fairness in predicting successful attackers. These practices reflect our commitment to ethical data handling and align with GDPR regulations and Ethical Guidelines for Statistical Practice.⁵

4.3 Potential Ethical Problems of NAC Breda and Recommendations for Improvement

The ethical issue at NAC Breda involves fan misconduct during matches, including incidents like throwing objects onto the field. This behaviour jeopardizes safety, tarnishes the club's reputation, and violates its values. Responsibility for managing fan conduct lies with various parties, including NAC Breda, law enforcement, and visiting/home teams, but the club must take the lead in addressing and preventing such incidents to avoid injuries and reputational damage.²

Recommendations include implementing strict codes of conduct, enhancing security measures, and enforcing regulations outlined in event terms and conditions to deter misconduct through penalties.

5 Recommendations

Based on our findings, we recommend NAC Breda prioritize acquiring young players aged 20-25 for their superior performance in goals, dribbles, and successful attacking actions. Considering players aged 25-30 can provide a balance between talent and financial feasibility. Utilizing the developed XGBoost machine learning model, focusing on key indicators like dribbles per 90, can enhance recruitment accuracy. With a predictive accuracy rate of 91% for successful attacking actions per 90, the model demonstrates its effectiveness in forecasting player suitability. This strategy can extend to other positions by identifying relevant performance metrics. Overall, by leveraging predictive analytics and prioritizing young talent, NAC Breda can optimize recruitment and strengthen its team for future success.²

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